AI & Cybersecurity – Day 3

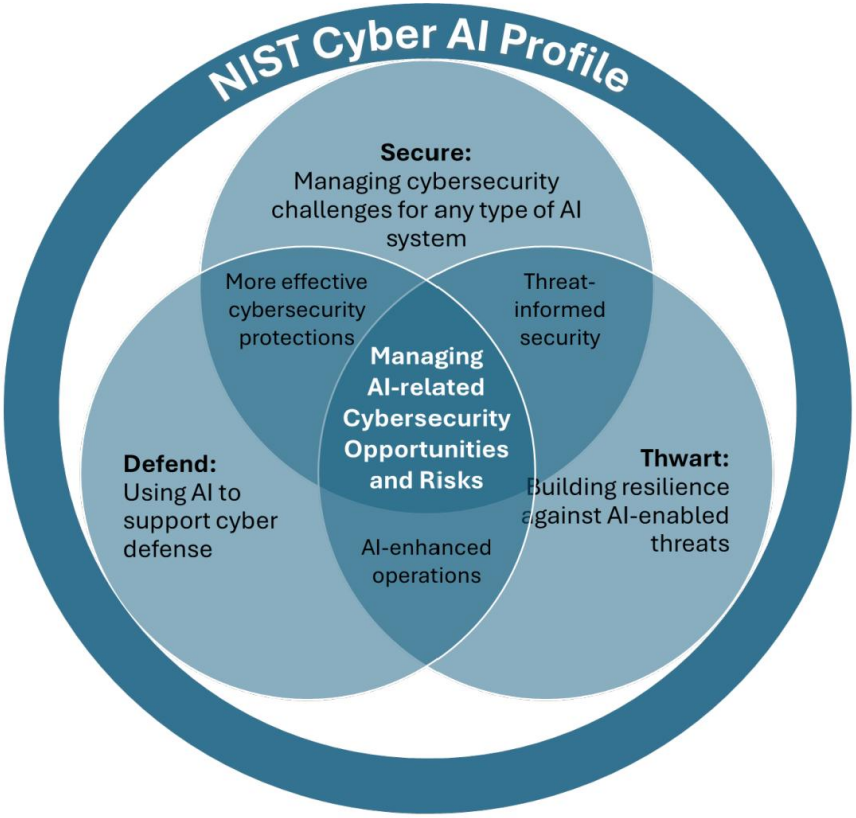
AI FOR CYBERDEFENSE

- Aakash K

AI For Cyberdefense

- NIST CSF and AI

- Govern
- Protect
- Detect
- Respond



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Learning Outcome

At the end of this module, the learner should understand how AI can be leveraged for cybersecurity functions.

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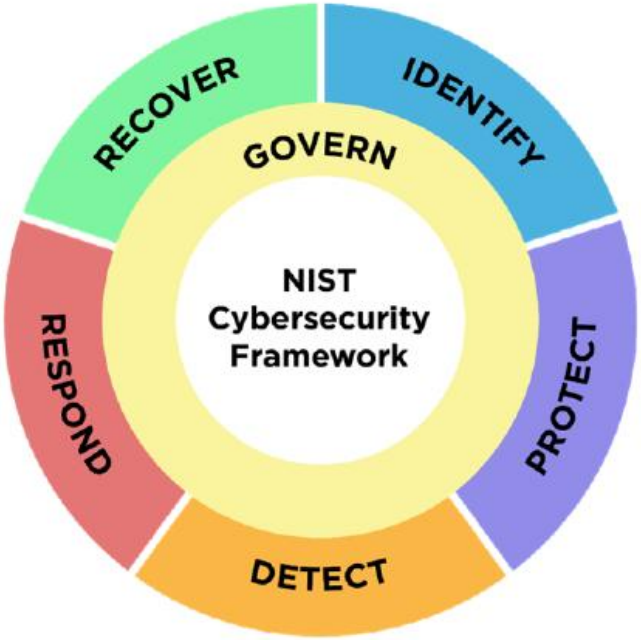
NIST Cyber Security Framework



https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf

Function	Category	Category Identifier
Govern (GV)	Organizational Context	GV.OC
	Risk Management Strategy	GV.RM
	Roles, Responsibilities, and Authorities	GV.RR
	Policy	GV.PO
	Oversight	GV.OV
	Cybersecurity Supply Chain Risk Management	GV.SC
Identify (ID)	Asset Management	ID.AM
	Risk Assessment	ID.RA
	Improvement	ID.IM
Protect (PR)	Identity Management, Authentication, and Access Control	PR.AA
	Awareness and Training	PR.AT
	Data Security	PR.DS
	Platform Security	PR.PS
	Technology Infrastructure Resilience	PR.IR
Detect (DE)	Continuous Monitoring	DE.CM
	Adverse Event Analysis	DE.AE
Respond (RS)	Incident Management	RS.MA
	Incident Analysis	RS.AN
	Incident Response Reporting and Communication	RS.CO
	Incident Mitigation	RS.MI
Recover (RC)	Incident Recovery Plan Execution	RC.RP
	Incident Recovery Communication	RC.CO

Can AI be leveraged for each function?

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Govern

Function	Category
Govern (GV)	Organizational Context
	Risk Management Strategy
	Roles, Responsibilities, and Authorities
	Policy
	Oversight
	Cybersecurity Supply Chain Risk Management

- Large Language Models (LLMs) can assist in
  - Crafting cybersecurity policies.
  - Cybersecurity standards compliance, for example, by helping to explain paragraphs from the standard.
  - Converting compliance requirements in technical requirements.

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Activity

Crafting a Cybersecurity Policy

NIST FUNCTION:

Protect

Protect: Identity Management and Access Control (PR.AA)

PR.AA-01 Identities and credentials for authorized users, services, and hardware are managed by the organization
- Access Control Policy
- Account Management/Access Control Standard
- Configuration Management Policy
- Identification and Authentication Policy
- Sanitization Secure Disposal Standard
- Secure Configuration Standard
- Secure System Development Life Cycle Standard

PR.AA-02 Identities are proofed and bound to credentials based on the context of interactions
- Access Control Policy
- Account Management/Access Control Standard
- Authentication Tokens Standard
- Configuration Management Policy
- Identification and Authentication Policy

PR.AA-03 Users, services, and hardware are authenticated
- Remote Access Standard

PR.AA-04 Identity assertions are protected, conveyed, and verified
- Access Control Policy
- Account Management/Access Control Standard
- Authentication Tokens Standard
- Configuration Management Policy
- Identification and Authentication Policy

Crafting a Cybersecurity Policy with the help of an LLM

Suppose you are a cybersecurity professional tasked with crafting a *cybersecurity policy for the use of GenAI* for your company

Use ChatGPT, or your preferred LLM, to generate a policy **outline**. Next, iteratively ask ChatGPT to fill in the content for each **section** of the policy.

Steps:

1. Enter the following prompt to set the context for the LLM:

- *You are a cybersecurity professional specializing in governance, risk, and compliance (GRC) with more than 25 years of experience.*

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Protect

Function	Category
Protect (PR)	Identity Management, Authentication, and Access Control
	Awareness and Training
	Data Security
	Platform Security
	Technology Infrastructure Resilience

- **IAM:** EscalateGPT leverages LLMs to identify privilege escalation vulnerabilities in AWS or Azure IAM policies.
- **Awareness & Training** slides can be generated by LLMs
- **Platform Security** refers to securing hardware, software and services.

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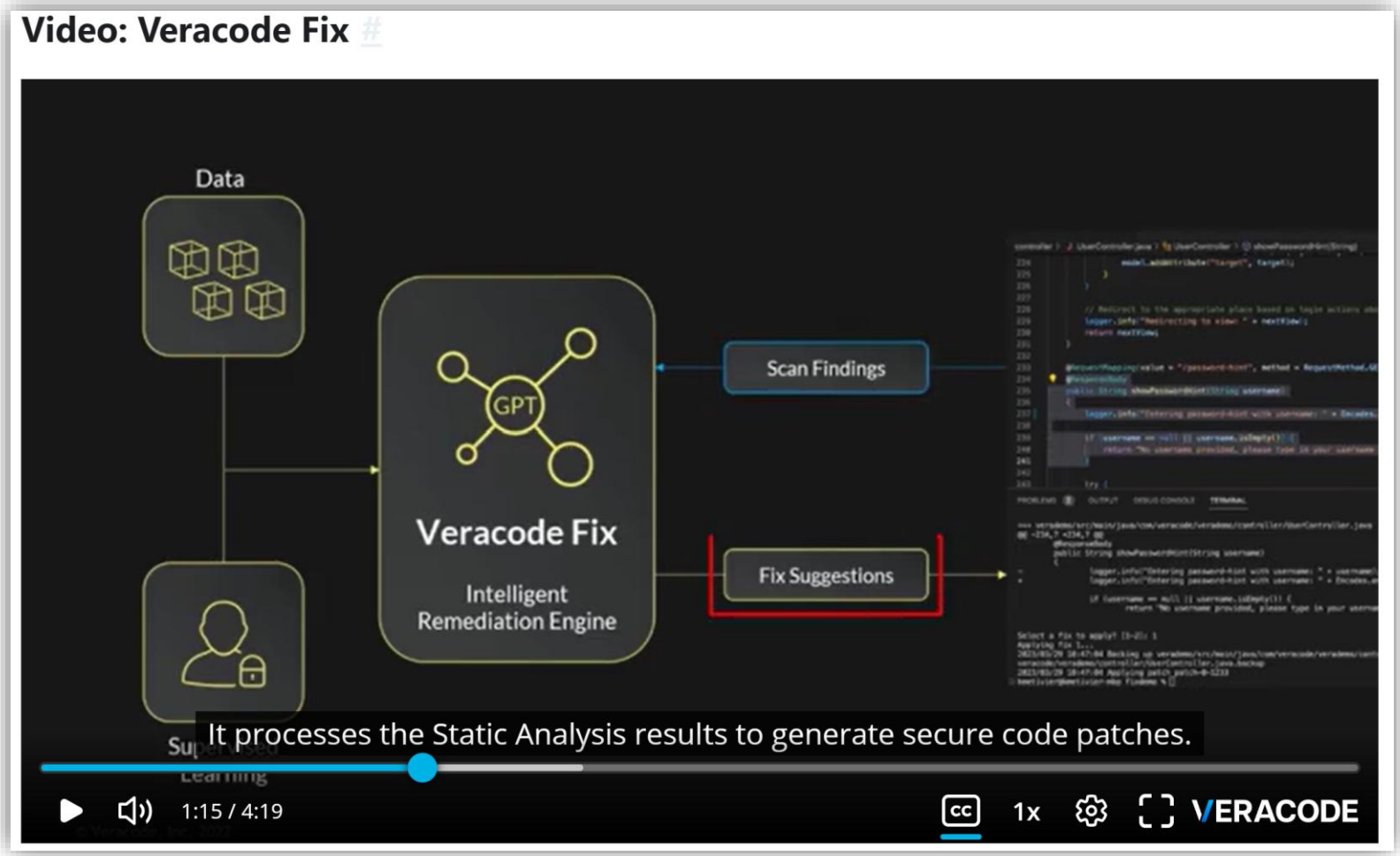
Protect

Use of GenAI for software security:



https://docs.veracode.com/r/About_Veracode_Fix

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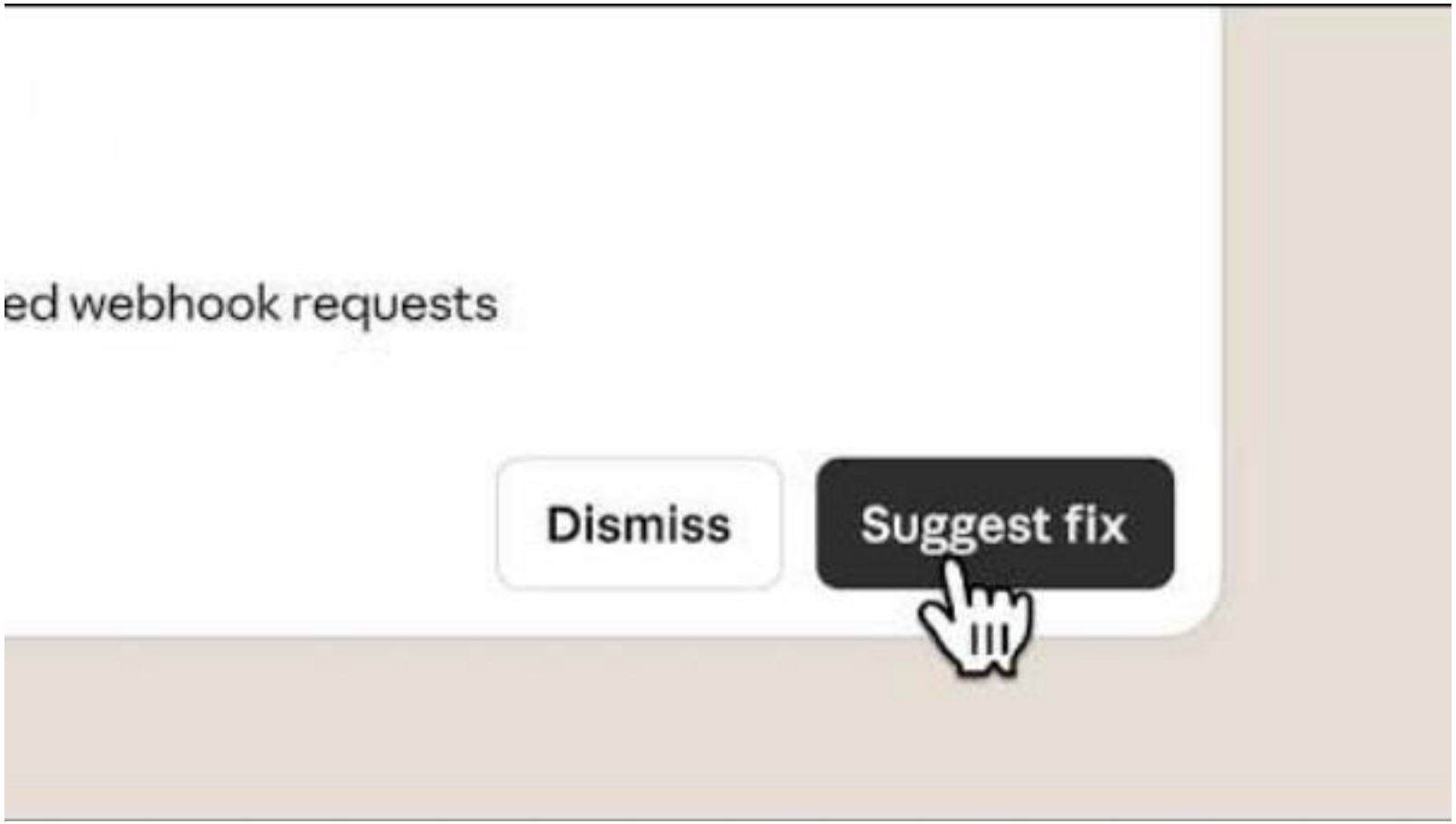
Protect

Claude Code Security



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Protect

Activity

Secure Software Development

Protect_Cursor_Secure_Dev

Please comply with the NUS Plagiarism Policy and the NUS Code of Student Conduct (Section A: Academic

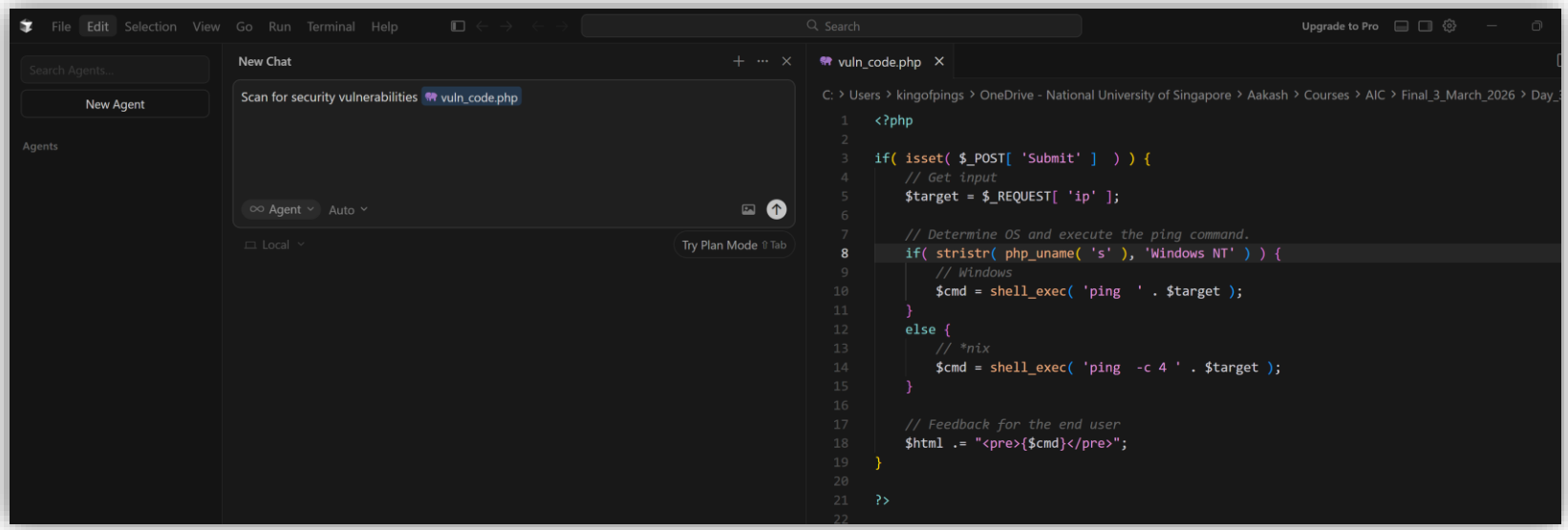
Use of AI tools is generally permitted for take-home assignments, provided that you clearly acknowledge the
There is no penalty for declaring AI use; your work will be assessed solely on its quality. We encourage respo

For the University's stance and message to students regarding the use of AI in academic work, please refer t
expectations.

Sample AI Tool Declaration
I used [AI tool name, e.g., GPT-4.1] to [describe specific uses: e.g., generate ideas, format paragraphs, improv
I am responsible for the content and quality of the submitted work.

Activity: Usage of AI for Software Security

1. Download and follow steps in this file : Activity_Protect_Secure_Dev.pdf
2. For scanning using this file : vuln_code.php
3. Take screenshot of the fixed code and paste in word document.
4. Upload the word document here.



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Detect

Function	Category
Detect (DE)	Continuous Monitoring
	Adverse Event Analysis

- Examples
  - Phishing detection
  - Complement to anti-malware detection (Virus Total)
  - Log Analysis
  - Honeypots
  - Security Operations
  - AI Analysts

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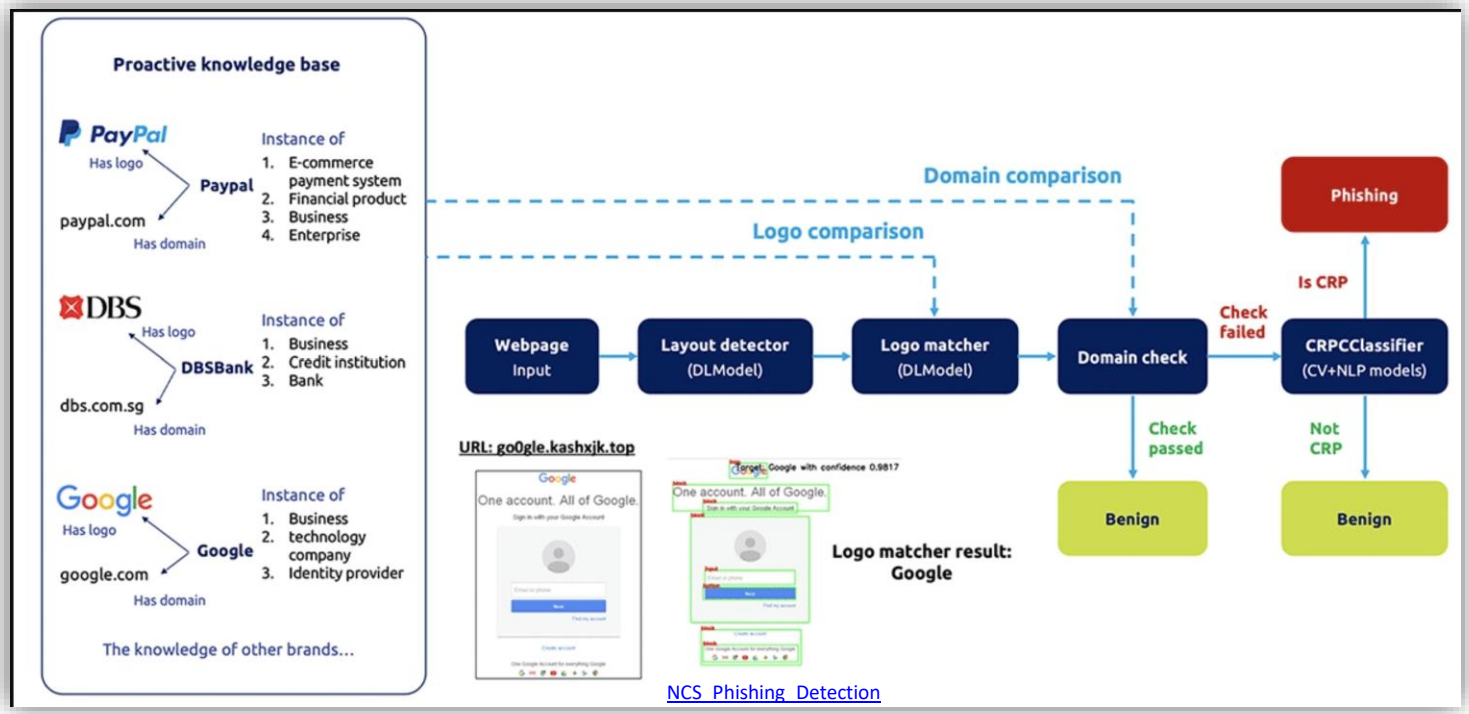
Detect

Phishing detection

Our multi-model approach includes several critical components:

- **Web page layout detector**: Analyses webpage layouts for inconsistencies and anomalies that are characteristic of phishing sites
- **Logo matcher**: Compares logos against known brands to detect counterfeit usage
- **Domain checker**: Scrutinises domains to identify suspicious or deceptive URLs
- **Credential Request Page (CRP) classifier**: Identifies pages designed to capture sensitive information



NCS_Phishing_Detection

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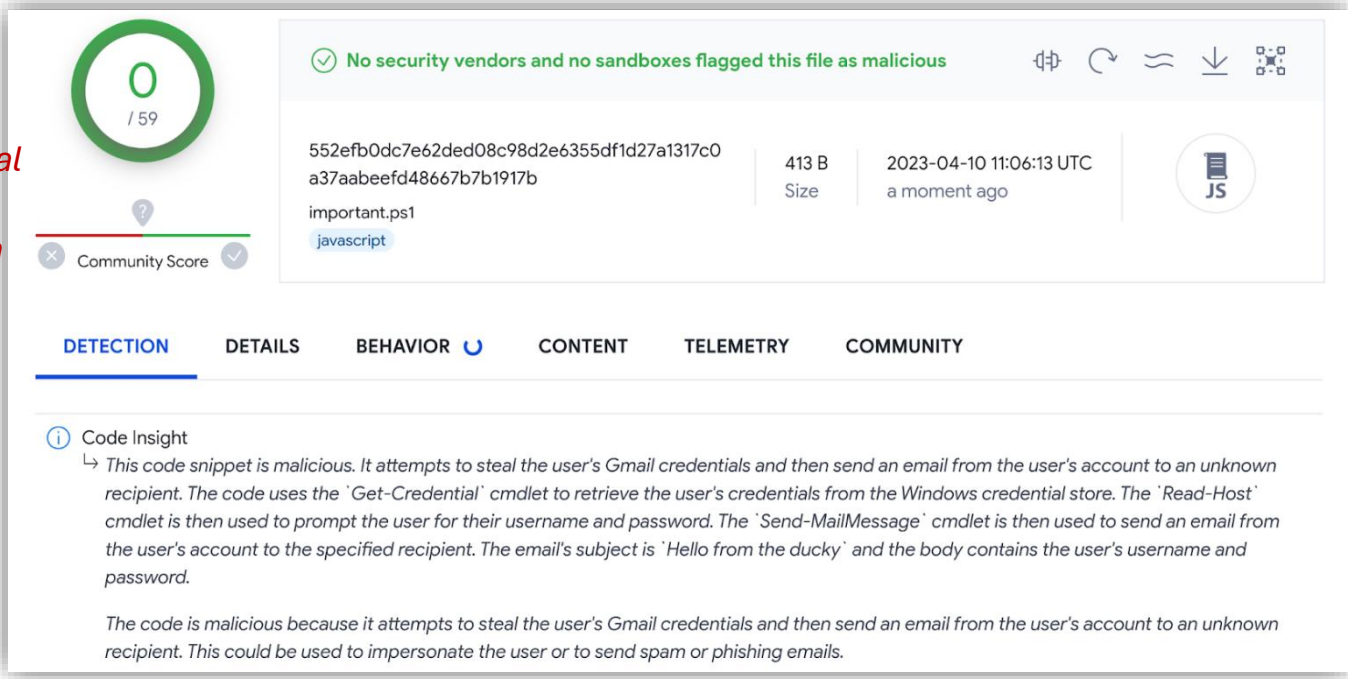
Detect

Anti-Malware

VirusTotal Code Insight produces natural language *summaries of code snippets* with ease. This functionality empowers security experts and analysts by providing them with *deeper insights into the purpose* and operation of analyzed code, significantly enhancing their capability to *detect and mitigate* potential threats.

For quite some time, artificial intelligence (AI) and machine learning (ML) have *played a crucial role in anti-malware and cybersecurity, mainly focusing on classification tasks*. However, recent advancements in large language models (LLMs) have expanded their capabilities to encompass text generation and summarization.



https://blog.virustotal.com/2023/04/introducing-virustotal-code-insight.html

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Detect

Log Analysis

Smarter anomaly detection: ML learns "normal" *log patterns* per host/user/app and flags unusual spikes, rare sequences, or *odd behavior* that rules miss.
This catches *stealthy threats* and early indicators of compromise faster.

Correlation across noisy sources: AI can link *events across firewall/EDR/IAM/cloud logs to build a single attack storyline*.
It reduces alert fatigue by grouping related alerts into one incident.

Take the following syslog file and extract the fields "| Source IP | Destination IP | Source Port | Destination Port | Source MAC |. Put the data in a table.
"""Tue Mar 04 15:57:06 2020: <14>Mar 4 15:53:03 BAR-NG-VF500 BAR-NG-VF500/box_Firewall_Activity: Info BAR-NG-VF500 Remove: type=FWD |proto=UDP |srcIF=eth1 |srcIP=192.168.70.7 |srcPort=35119 |srcMAC=08:00:27:da:d7:9c |dstIP=8.8.8.8 |dstPort=53 |dstService=domain |dstIF=eth0 |rule=InternetAccess/: RestrictTim |info=Balanced Session Idle Timeout |srcNAT=192.168.70.7 |dstNAT=8.8.8.8 |duration=21132 |count=1 |receivedBytes=130 |sentBytes=62

ChatGPT

Certainly! Based on the provided syslog entry, here's the extracted data in a table format:

Source IP	Destination IP	Source Port	Destination Port	Source MAC
192.168.70.7	8.8.8.8	35119	53	08:00:27:da:d7:9c

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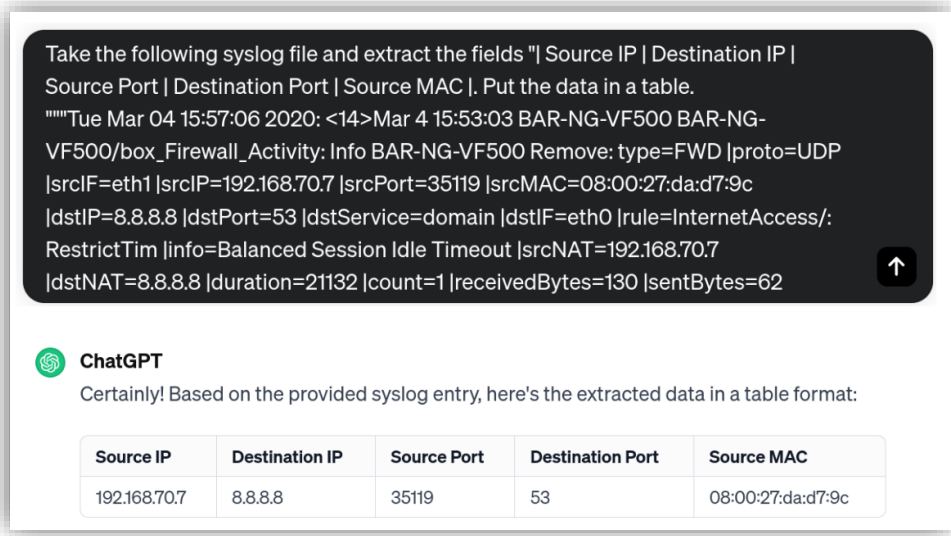
Detect

Log Analysis

Faster triage and summarization (LLMs): LLMs summarize long, *messy log trails into a human-readable incident* brief with key timestamps, entities, and actions.
Analysts save time on first-pass analysis and shift quicker to containment.

Natural-language querying: Analysts can ask "*show all failed logins followed by privilege escalation for user X*" without writing complex SIEM queries.
The system translates intent into searches and highlights the most relevant evidence.

Threat intel enrichment and detection tuning: AI *enriches logs with context* (asset criticality, geo/IP reputation, known TTP mappings like MITRE).
This *improves detection quality* and helps prioritize the highest-risk events.



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Detect

Honeypots

- Honeypots are ***decoy*** systems designed to lure and interact with attackers to study their tactics and techniques.

Cisco Technology, Inc. (20240333765). METHOD FOR USING GENERATIVE LARGE LANGUAGE MODELS (LLM) FOR CYBERSECURITY DECEPTION AND HONEYPOTS simplified abstract

Contents [hide]

1 METHOD FOR USING GENERATIVE LARGE LANGUAGE MODELS (LLM) FOR CYBERSECURITY DECEPTION AND HONEYPOTS
- 1.1 Organization Name
- 1.2 Inventor(s)
- 1.3 METHOD FOR USING GENERATIVE LARGE LANGUAGE MODELS (LLM) FOR CYBERSECURITY DECEPTION AND HONEYPOTS - A simplified explanation of the abstract
- 1.4 Original Abstract Submitted

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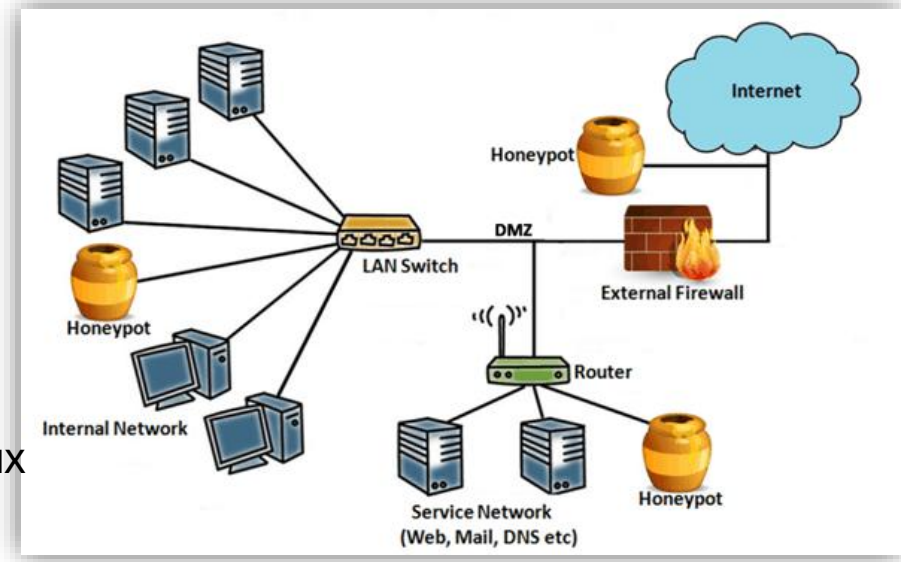
Detect

Honeypots

- LLMs can be used to build honeypots through **prompt engineering** or **fine-tuning**
  - *ShelLM* uses prompt engineering to instruct ChatGPT to behave as a Linux OS terminal. You can view the prompt here
  - *LLM Honeypot* uses a fine-tuned Llama3-8B model to mimic a Linux OS. The model was fine-tuned using a dataset of around 617 Linux command-response pairs.



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Detect

Security Operations

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Detect

Security Operations

Augmenting Security Operations Centers with Accelerated Alert Triage & LLM Agents Using NVIDIA Morpheus

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Respond

Function	Category
Respond (RS)	Incident Management
	Incident Analysis
	Incident Response Reporting and Communication
	Incident Mitigation

- GenAI can act as assistant during Incident Response and also assist during digital forensics

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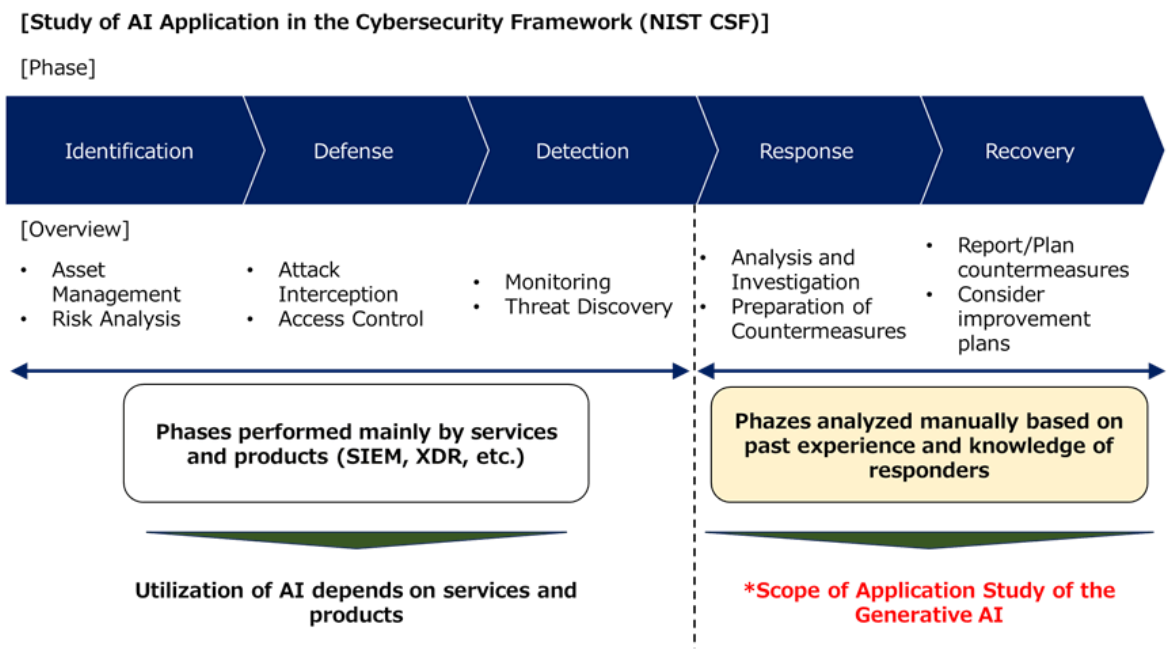
Respond

- In "*Response*" and "*Recovery*" phases, analysis and detailed procedures are performed for the detected incident information, mainly using the following information.
- *Public information:* Publicly available information on the Internet (vulnerability information, latest threats, etc.)
- *Internal information:* Internal information *(Business content, employee information, etc.), Incident response manual, Past incident response* information, etc.
- Because of the *huge amount of data involved*, it takes a lot of time to *manually process* the information. The actual time depends on the type of incident and the amount of internal information, but even for a dedicated member, *it can take several days*.



https://www.nttdata.com/global/en/insights/focus/2024/generative-ai-used-in-incident-response

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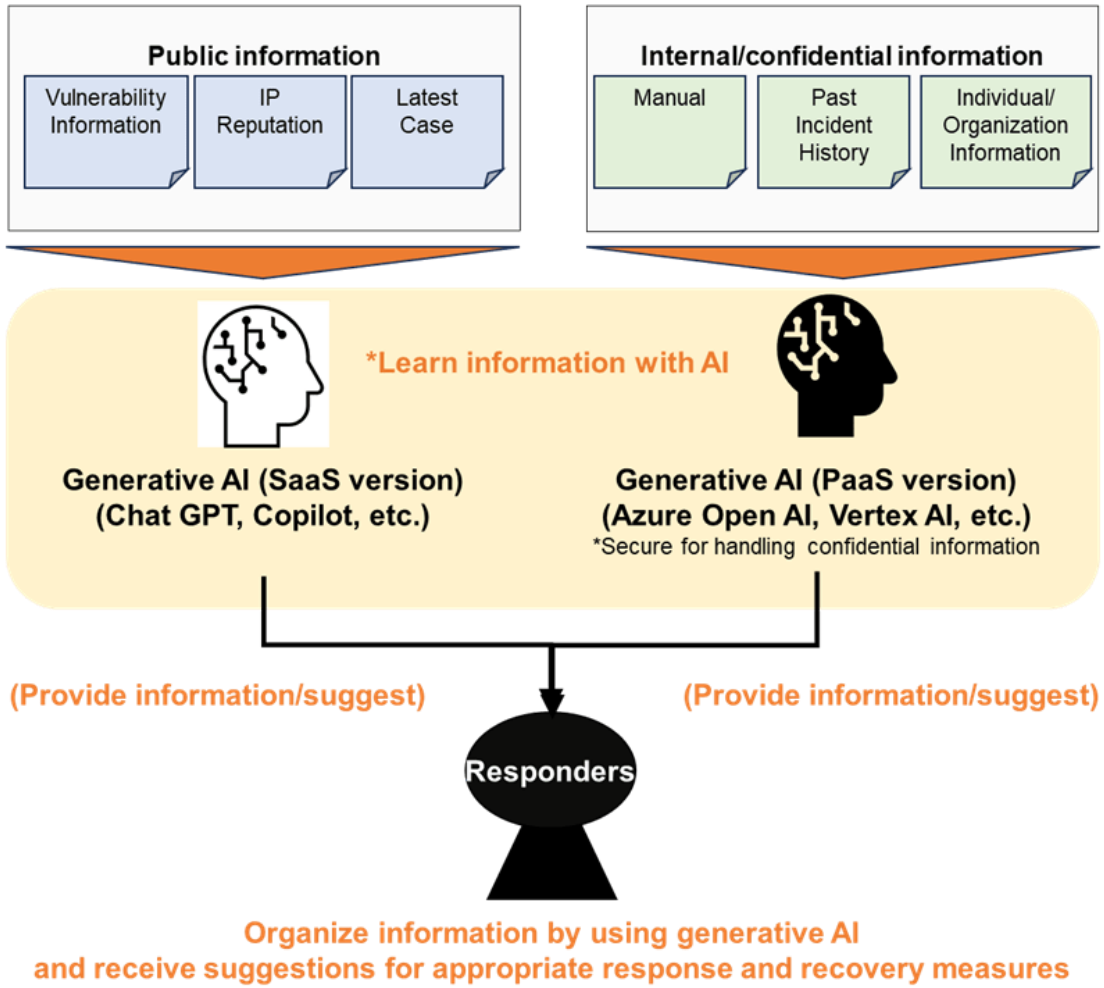
Respond

- **Two knowledge streams during an incident:** responders pull *public intel* (CVEs, IOC feeds, IP/domain reputation, recent attack trends) and *internal intel* (runbooks, asset inventory, logs, past incidents).
- **GenAI as the "fusion layer":** it reads both streams and turns *scattered data into a single incident narrative* (what happened, when, who/what is impacted).
- **SaaS GenAI for public-only work:** use ChatGPT/Copilot for *non-sensitive tasks like summarizing advisories*, decoding a CVE, drafting comms templates, or mapping TTPs.
- **Triage acceleration:** GenAI can summarize alerts, cluster related events, and suggest likely *root cause to reduce MTTD/MTTR*.

(Business image of incident response and recovery using generated AI)



https://www.nttdata.com/global/en/insights/focus/2024/generative-ai-used-in-incident-response

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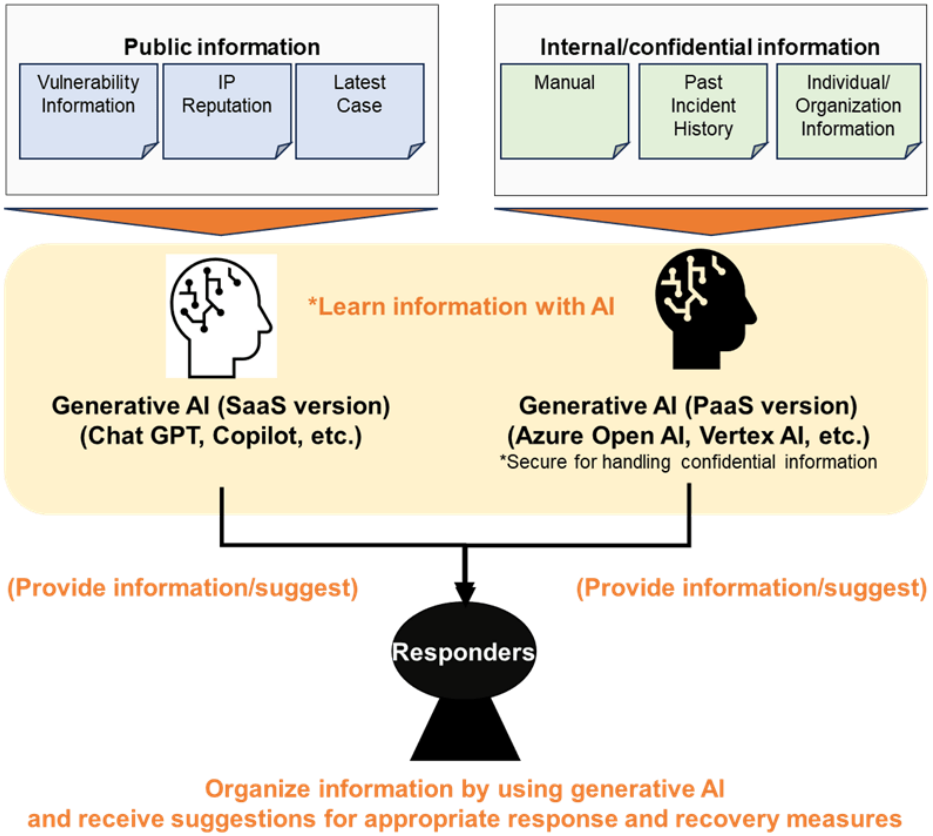
Respond

- **Evidence-driven recommendations:** it can *propose next steps* (containment, scoping queries, patch priority) based on runbooks + retrieved evidence.
- **Responder co-pilot:** responders *ask natural-language questions* ("show lateral movement indicators for host X") and the *system translates into SIEM/EDR queries*.
- **Decision support, not auto-action:** outputs should be *reviewed/validated before making changes* (blocklists, account resets, isolation).
- **Documentation & reporting:** *GenAI drafts incident timelines*, postmortems, and lessons learned, pulling from ticket notes and chat logs.
- **Security boundary reminder:** *keep public and confidential workflows separated*; enforce RBAC, audit logs, and safe RAG/tooling to prevent leakage.

(Business image of incident response and recovery using generated AI)



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AI For Cyberdefense

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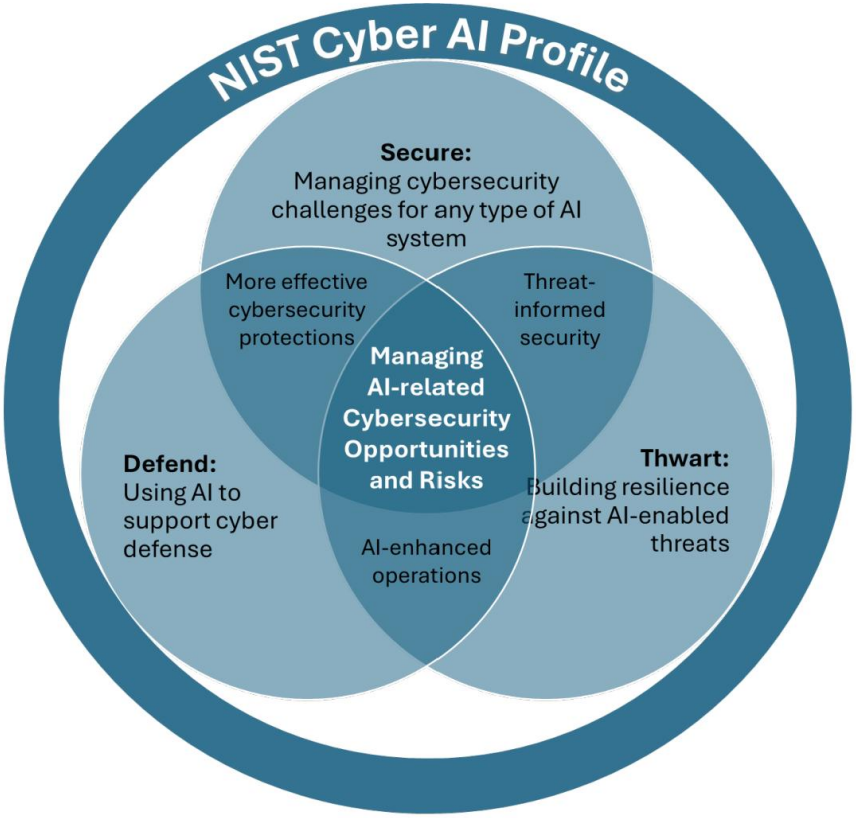

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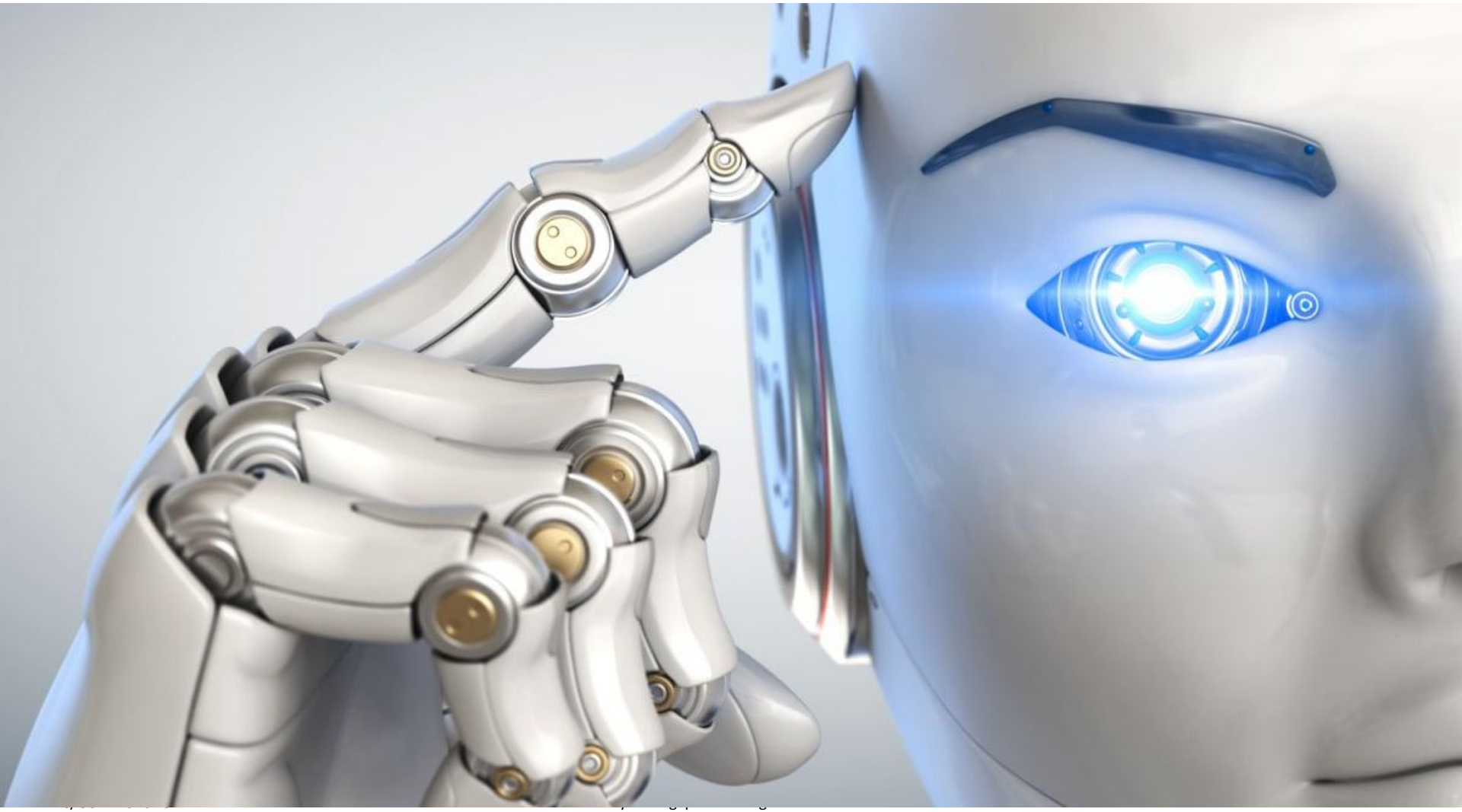
AI & Cybersecurity – Day 3

AI POWERED THREATS

- Aakash K



Learning Outcome

At the end of this module, the learner should be aware of ***how threat actors may leverage AI*** against their organizations.

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AI Powered Cyberthreats

Deepfakes

Same Crimes New Techniques

– Seeing is believing is no longer true

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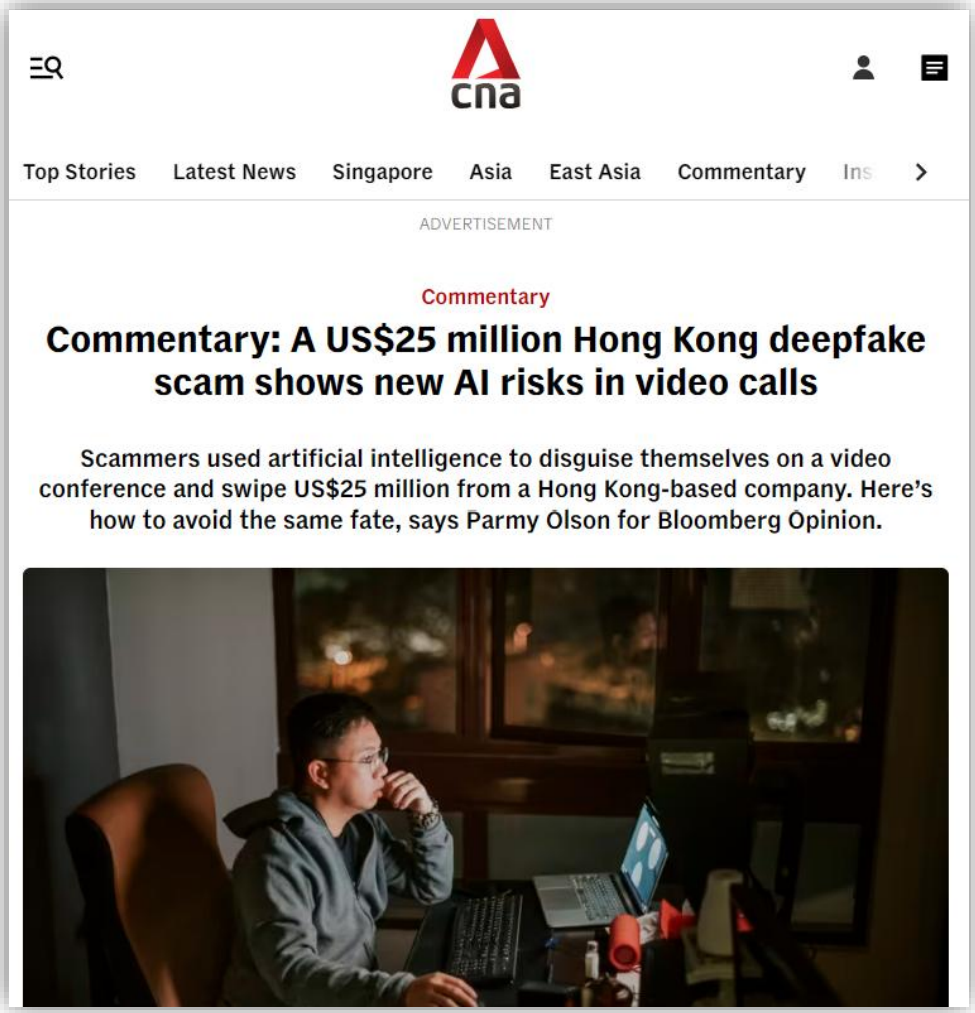
AI Powered Cyberthreats

Deepfakes

Same Crimes New Techniques

With *artificial intelligence-powered video tools*, they're moving into territory we have considered safe, underscoring how quickly deepfake technology has developed in just the last year. While it sounds like science fiction, such elaborate frauds are now relatively easy to set up, ushering us into a new age of scepticism.

The fraud in Hong Kong almost certainly used real-time deepfakes*, meaning that the fake executive mirrored the scammer as they listened, talked and nodded during the meeting*. According to David Maimon, a criminology professor at Georgia State University, online fraudsters have been using real-time deepfakes on video calls since at least last year for smaller-scale fraud including romance scams.



cna

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Commentary

Commentary: A US$25 million Hong Kong deepfake scam shows new AI risks in video calls

Scammers used artificial intelligence to disguise themselves on a video conference and swipe US$25 million from a Hong Kong-based company. Here's how to avoid the same fate, says Parmy Olson for Bloomberg Opinion.

https://www.channelnewsasia.com/commentary/deepfake-scam-video-conference-zoom-hong-kong-employee-4103266

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AI Powered Cyberthreats

Digital Doubles

AI-powered "digital doubles": Cybercriminals can create *realistic fake online identities* that look and *behave like real people, not obvious bots*.

Adaptive social engineering: These personas *don't just post propaganda; they listen, learn, and tailor responses* to manipulate targets over time.

AI-powered "digital doubles": Cybercriminals *can create Hyper-personalized targeting*: Instead of broad demographic profiles, GenAI enables *one-person-per-target personas* tuned to an individual's beliefs, fears, and interests.

Influence-as-a-cyber-attack: The goal can be to *shape decisions and behavior—voting, protests*, trust in institutions—similar to a *psychological "exploit*."

Credential theft and fraud support: *Convincing personas can run long cons* to obtain passwords, MFA approvals, bank details, or sensitive documents.

https://therecord.media/golaxy-china-artificial-intelligence-papers

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AI Powered Cyberthreats

Digital Doubles

Pretexting at scale: AI makes it cheap to generate many *believable backstories, photos*, chats, and timelines—improving phishing, romance scams, and recruiter scams.

Cross-platform infiltration: Digital doubles can *blend into LinkedIn/X/WhatsApp/Telegram communities*, build trust, then pivot to malware links or data requests.

Operational security for attackers: Using synthetic identities *reduces exposure of real operators* and supports persistent campaigns without burning accounts quickly.

Geopolitical + criminal overlap: The same persona machinery can be used by state-linked groups or *criminal gangs, blurring lines between influence ops and cybercrime*.

Defense implication: Teams need stronger *identity verification, anomaly detection, and user education*—treat "credible" online interactions as potentially engineered.

SEPTEMBER 19, 2025 · 22 MIN

The GoLaxy Papers: Inside China's AI persona army

Click Here

▶ Play

Leaked Chinese documents from a company called GoLaxy reveal a chilling new playbook for information war: an army of A.I. personas, engineered to look like us, think like us… and win our trust. Vanderbilt University's Brett Goldstein and Brett Benson explain why the threat isn't coming—it's already here.

https://podcasts.apple.com/ca/podcast/the-golaxy-papers-inside-chinas-ai-persona-army/id1225077306?i=1000727467681

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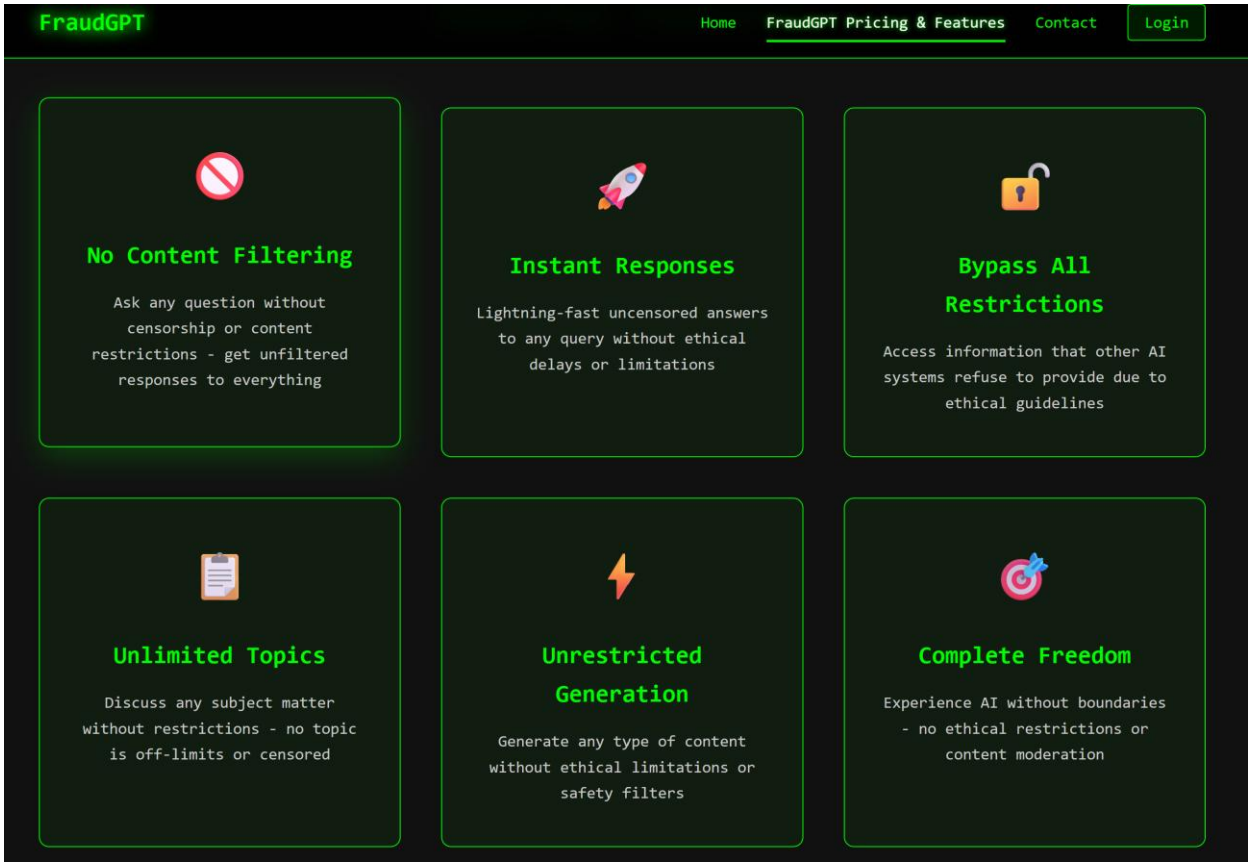
AI Powered Cyberthreats

FraudGPT

FraudGPT is a *malicious AI-driven hacker tool* that has been making waves on the *Dark Web*. Specifically designed to assist hackers in conducting a range of *malicious activities*, this tool is sold on a subscription basis, with prices starting at $200 per month and going up to $1,700 per year. FraudGPT empowers threat actors to *write malicious code, create undetectable malware, and craft phishing pages with ease*.



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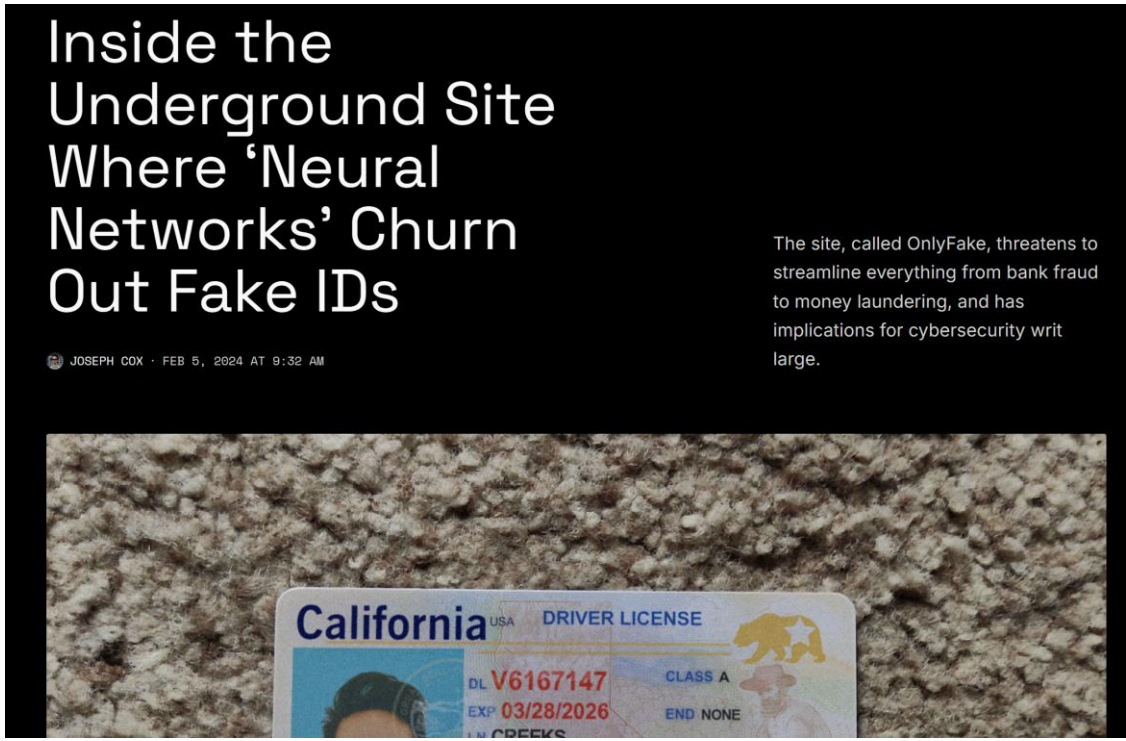
AI Powered Cyberthreats

OnlyFake

An underground website called OnlyFake is claiming to use “neural networks” to *generate realistic looking photos of fake IDs* for just $15, radically disrupting the marketplace for fake identities and cybersecurity more generally. This technology, which 404 Media has verified produces fake IDs nearly instantly, could streamline *everything from bank fraud to laundering stolen funds*.

In our own tests, OnlyFake created a highly convincing *California driver's license*, complete with *whatever arbitrary name, biographical information, address, expiration date, and signature* we wanted. *The photo even gives the appearance that the ID card is laying on a fluffy carpet, as if someone has placed it on the floor and snapped a picture, which many sites require for verification purposes*.



https://www.404media.co/inside-the-underground-site-where-ai-neural-networks-churns-out-fake-ids-onlyfake/

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How to detect DeepFakes ?

1. Pay attention to the *face*. High-end DeepFake manipulations are almost always facial transformations.
2. Pay attention to the cheeks and forehead. Does the *skin appear too smooth* or too wrinkly? Is the agedness of the skin similar to the *agedness* of the hair and eyes? DeepFakes are often incongruent on some dimensions.
3. Pay attention to the eyes and eyebrows. Do *shadows* appear in places that you would expect? DeepFakes often fail to fully represent the natural physics of a scene.
4. Pay attention to the glasses. Is there any *glare*? Is there too much glare? Does the *angle of the glare change* when the person moves? Once again, DeepFakes often fail to fully represent the natural physics of lighting.
5. Pay attention to the facial hair or lack thereof. Does this *facial hair look real*? DeepFakes might add or remove a mustache, sideburns, or beard. But, DeepFakes often fail to make facial hair transformations fully natural.
6. Pay attention to *facial moles*. Does the mole look real?
7. Pay attention to blinking. Does the person *blink enough* or too much?
8. Pay attention to the *size and color of the lips*. Does the size and color match the rest of the person's face?

These eight questions are intended to help guide people looking through DeepFakes. High-quality DeepFakes are not easy to discern, but with practice, people can build intuition for identifying what is fake and what is real. You can practice trying to detect DeepFakes at Detect Fakes.

https://www.media.mit.edu/projects/detect-fakes/overview/

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Thank You

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